EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Step Over a method call means it is not executed.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# OOP debugging with IDEs

Name: _____ Class: _____ Date: _____

**1. Which structure supports BFS traversal?**

**2. What is an object's field?**

**3. When a local variable shadows a field, which name is used without qualification?**

**4. Why might Step Into be more useful than Step Over for a faulty function?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A FIFO queue.

### 2. Model answer

Stored state belonging to that object.

### 3. Model answer

The local variable.

### 4. Model answer

It lets you inspect statements, parameters and field updates inside the called function.

### Vocabulary — know these terms

breakpoint — A chosen instruction where execution pauses.

Step Into — Enter a called function for debugging.

Step Over — Execute a call without entering it line by line.

watch — IDE display of chosen variables/expressions.

object state — Current values of an object's fields and properties.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Design an IDE-debugging workflow for a temperature sensor whose Readings list gains two entries after a single AddReading() call.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Explain Step Into and Step Over at a method call.

### Question 2

A class method Increase(int n) contains `int Total = Total + n;` while Total is an existing property. Identify the issue and write correct C# code.





NEW LEARNING

# Read, annotate and explain

### Set a breakpoint

Pause at a method call or the following instruction to inspect current state.

### Step into to inspect internals

Step Into executes the called method one instruction at a time. Watch Count and parameter n.

### Step over to skip internals

Step Over runs the entire call and pauses at the next statement, allowing you to inspect the result.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.**

### Predict first

Step Into Add and inspect the local points and the field Points.

### Model the reasoning

The local becomes 8 while the object field remains 3; the assignment targets the wrong variable.

### Model the reasoning

Replace it with Points += amount; and rerun. The field should be 8.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve